

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	examine how people use data to develop explanations and scientific knowledge to meet needs or solve problems
KEY IDEA	Scientists organise observations as data, look for patterns, develop explanations and test whether a solution meets a need. Evidence can support an explanation without making it certain forever.
WORKED MODEL	Repeated school temperature readings show the unshaded area is consistently warmer. The data supports adding shade, then new measurements test whether the solution works.

SUCCESS CRITERIA - I CAN

☐ separate data and opinion ☐ explain a pattern ☐ connect evidence to a solution ☐ recognise limits

A. BUILD THE SCIENCE IDEA

1. Label each as observation, data, explanation or solution: 31 C; the surface felt hot; dark surfaces absorb more energy; add a shade sail.

2. Explain why repeated readings are stronger than one reading.

3. Identify the need, the evidence and the proposed solution in the worked model.

B. APPLY AND EXPLAIN

4. A class records plant growth in three soils. Describe one pattern the table could show.

5. Write an explanation that connects the pattern to a science idea without saying proved.

6. List two criteria for deciding whether a proposed solution is successful.

C. INDEPENDENT CHECK

7. Create a flow chart: question -> data -> pattern -> explanation -> solution -> new data.

8. Check where a claim could be weakened by missing or unfair data.

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D. APPLY SCIENCE IN CONTEXT

9. A town tests three water filters. Filter A removes 70%, B 88%, C 91% of particles; C costs four times more than B.

10. Recommend a filter using the data plus cost, access and safety considerations.

11. State what further evidence is needed before the town decides.

E. REASON, EVALUATE AND TRANSFER

12. Evaluate: The highest result automatically gives the best solution.

13. Write a short community explanation that separates finding, uncertainty and recommendation.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can use data to develop explanations and evaluate science-based solutions.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. data; observation; explanation; solution.
2. Repetition shows consistency and reduces the influence of unusual results.
3. Need: cooler area; evidence: repeated temperatures; solution: shade.
- 4-6. A valid pattern uses comparable data; the explanation links evidence to science; criteria can include effectiveness, safety, cost and access.
7. All six stages should be linked; new data evaluates the solution.
8. Missing controls, small sample, inconsistent measures or selected data weaken claims.
9. B and C perform similarly; cost differs greatly.
10. B may be defensible if safe because performance is close to C at lower cost; accept other evidence-based choices.
11. Repeated trials, contaminant types, maintenance, lifespan, safety and full cost.
12. False; effectiveness is one criterion and data quality matters.
13. Must name data, uncertainty/trade-off and a justified recommendation.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

Data does not speak for itself; an explanation must connect the pattern to a science idea.

Science informs decisions, but cost, access, safety and community needs also matter.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.